Dehazing Echocardiography Challenge 2025

A Technical Report of Team TU Munich

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Abstract. In this technical report, we present our contributions to the Dehazing Echocardiography Challenge 2025, focusing on improving the quality of echocardiographic images degraded by acoustic haze. Inspired by advances in generative modeling, particularly the M2O-DiffGAN framework, we propose two complementary generative-model-based pipelines. The first employs a CycleGAN to directly translate hazy ultrasound images into dehazed counterparts without requiring paired training data. The second builds upon the CycleGAN output by training an Enhanced Diffusion for Super Resolution (EDiffSR) model on synthetic paired images, enabling iterative refinement of haze removal while preserving anatomical structures. Both pipelines integrate a U-Net ResNet34 segmentation model to identify regions of interest, facilitating targeted contrast enhancement via a generalized Contrast-to-Noise Ratio (gCNR) optimization step that improves ventricle visibility while preserving the ventricle septum. Evaluation on the challenge dataset demonstrates that both approaches effectively suppress haze, enhance structural contrast, and maintain tissue fidelity, with the diffusion-based pipeline providing slightly better performance according to the evaluation metrics at higher computational complexity. These results highlight the potential of generative models, including adversarial and diffusion-based frameworks, to improve diagnostic reliability in challenging echocardiographic imaging scenarios.

Keywords: Dehazing Ultrasound · DehazingEcho2025 · CycleGAN · Diffusion.

1 Introduction

Echocardiography is one of the most widely used imaging modalities for the diagnosis and monitoring of cardiac disease due to its real-time capabilities, cost-effectiveness, and non-invasive nature. However, the diagnostic value of echocardiac images is often limited by poor image quality. Acoustic clutter and haze arise from multipath reflections caused by layers of skin, fat, and intercostal muscle between the transducer and the heart. These structured noise artifacts obscure important anatomical structures and degrade contrast, making accurate clinical interpretation difficult. The problem is particularly pronounced in patients

with challenging acoustic windows, such as those with obesity, where standard B-mode imaging may be insufficient. Although techniques such as tissue harmonic imaging have been developed to reduce clutter, they remain susceptible to haze in severe cases. Conventional denoising algorithms, while effective against uncorrelated noise, typically fail to address the complex and highly structured statistical nature of haze.

Recent advances in generative models have opened new avenues for medical image restoration. Diffusion-based methods, in particular, have shown remarkable success in inverse imaging problems by progressively refining noisy signals through stochastic sampling. Likewise, adversarial networks such as CycleGANs have demonstrated strong performance in domain adaptation and image-to-image translation tasks, offering an alternative strategy for mapping degraded images to clean counterparts without requiring paired training data. These approaches are especially well-suited for ultrasound imaging, where acquiring clean ground-truth references is inherently difficult.

In this work, we present two complementary generative-model-based pipelines for dehazing echocardiac images. The first approach leverages a CycleGAN-based generator, which directly translates hazy ultrasound images into dehazed counterparts in a single forward pass. Building upon this, the second approach employs an Enhanced Diffusion for Super Resolution (EDiffSR)[5] framework, a conditional diffusion model that is trained on synthetic paired data generated by the CycleGAN. By using these CycleGAN-derived fake pairs, the diffusion model learns to gradually remove haze through stochastic differential equations, enabling more refined dehazing while preserving anatomical detail. In both cases, segmentation with a U-Net ResNet34 [4] model is used to delineate regions of interest, and a gCNR-optimized enhancement step ensures that the ventricle region is enhanced while the ventricle septum region is preserved. Using the evaluation metrics on the challenge dataset, we demonstrate that our generative frameworks effectively suppress haze while enhancing contrast, offering a promising pathway toward improved diagnostic reliability in challenging echocardiographic imaging scenarios.

2 Related Work

2.1 Multi-source Domain Transformation

Recent work by Huang et al. [2] introduced M2O-DiffGAN, a novel approach for standardizing ultrasound images across multiple medical centers. This method addresses the critical challenge of multi-source domain transformation (MSDT) by combining many-to-one adversarial learning with conditional adversarial diffusion models [6].

The key innovation of M2O-DiffGAN is its ability to jointly model multiple unlabeled ultrasound domains without requiring paired training data. The framework consists of a unified forward GAN and multiple domain-specific backward GANs

that work together to produce paired target-source samples, which then guide a fast conditional adversarial diffusion module. To address the unique challenges of ultrasound imaging, such as limited perceptual information and low resolution, the method incorporates an ultrasound-specific content loss that enhances feature preservation during transformation.

Validated on six clinical datasets covering thyroid, carotid, and breast ultrasound images, M2O-DiffGAN demonstrates substantial improvements over existing methods, achieving gains of 0.390, 0.120, 0.245, and 0.250 in MI, Bhattacharyya Coefficient, Dice, and IoU metrics respectively. Importantly, the method enables downstream segmentation models trained on target domains to generalize effectively to transformed source images, demonstrating its practical value for deploying AI diagnostic tools across diverse clinical settings with varying imaging protocols and equipment.

In the following section 3, we extend the M2O-DiffGAN framework to the task of removing structured haze from ultrasound echocardiography images. By tailoring the model to this application, we aim to substantially improve image quality and enhance the interpretability of clinically relevant features.

3 Method

We propose a method that adapts the M2O-DiffGAN [2] framework by focusing on one-to-one image translation rather than many-to-one. Specifically, we employ a CycleGAN architecture to translate hazy echocardiography images into clean counterparts and vice versa. The resulting cleaned images are subsequently refined using a diffusion-based enhancement module. Finally, localized gCNR enhancement is applied to the septum and ventricular regions to further improve image clarity and diagnostic interpretability.

3.1 CycleGAN

The CycleGAN-based generator serves as the first dehazing component. It performs image-to-image translation from hazy to clean ultrasound domains without requiring paired training data. The model was trained for 100 epochs on the challenge dataset, employing extensive data augmentation such as stretching, rotation, and brightness adjustments to improve robustness.

The architecture is a ResNet-based generator with nine residual blocks. During inference, a single forward pass through the generator (`G_BA`) produces a dehazed output. This direct translation avoids iterative refinement and provides computational efficiency compared to diffusion models. In [2], CycleGAN is presented as the initial step to synthesize paired data, which is then used to train a diffusion model in a supervised manner for domain transformation. For our problem of dehazing ultrasound images we also use a CycleGAN to generate synthetic image pairs, but also evaluate it as a standalone method, as it already yielded promising results on its own.

3.2 Diffusion

Following the M2O-DiffGAN approach, the diffusion-based component builds upon the CycleGAN output. Specifically, the diffusion model (EDiffSR) [5] was trained on synthetic image pairs generated by the CycleGAN: fake hazy images paired with their clean counterparts. This provides the supervised signal required for diffusion-based training in a setting where real paired data are unavailable.

EDiffSR is based on the ConditionalNAFNet architecture and employs stochastic differential equations (SDEs) to iteratively refine images. With 100 reverse steps and a cosine noise schedule, the model gradually removes haze to reconstruct a clean grayscale image. This approach has the potential to offer a more gradual and fine-grained restoration compared to the direct CycleGAN translation.

3.3 Postprocessing

After dehazing, both the CycleGAN and Diffusion pipelines undergo a few post-processing steps to enhance the results further. These processes are used for both pipelines, with a specific blending technique applied only in the Diffusion approach.

Segmentation In both the CycleGAN and Diffusion pipelines, post-processing begins with segmentation. A U-Net ResNet34 model is used to classify each pixel into one of three classes: background, ROI-1 (ventricle septum), and ROI-2 (ventricle). The model was trained for 200 epochs on the given pairs of hazy images and ROI-masks. The segmentation process enables the selective enhancement of the ventricle region, ensuring that the ventricle septum is preserved while improving the visibility of the ventricle.

gCNR Enhancement Following segmentation, the gCNR-based post-processing enhancement is applied in both pipelines. The ROI-1 region, corresponding to the ventricle septum, is preserved, while the ROI-2 region (the ventricle) undergoes gamma correction and smooth blending. This step optimizes the contrast-to-noise ratio and removes white noise from the ventricle region. Gaussian feathering ($\sigma = 3$) is applied at the boundaries of the regions to ensure smooth, natural transitions. This localized enhancement improves the contrast in the ventricle region while keeping the ventricle septum structure intact.

Blending Blending is performed in both approaches, but with slight differences in technique. For both the CycleGAN and Diffusion models, the inital ROI-1 of the hazy image is merged back on the dehazed image using ROI-based alpha blending. This ensures that the structure of the ventricle septum is preserved.

For the Diffusion approach, additional blending is performed to merge the upper and lower parts of the dehazed image. The blending region consists of 70% of

the dehazed image from the top and 30% from the bottom of the original hazy image, with 40-pixel feathering applied to create a smooth transition. This extra step is particularly beneficial for achieving a lower FID, as the Diffusion model tends to apply more significant changes to the lower part of the image, where the heart pulse is located.

3.4 Approach Summary

The dehazing pipelines based on CycleGAN and Diffusion share several common postprocessing steps but differ in their primary dehazing methods and blending strategies.

CycleGAN Approach The CycleGAN-based pipeline operates as follows:

1. CycleGAN generator performs direct image-to-image translation to produce dehazed outputs.
2. The dehazed image is segmented using U-Net ResNet34 to obtain ROI masks.
3. gCNR enhancement is applied to ROI-2 (ventricle), preserving ROI-1 (ventricle septum).
4. ROI-1 is merged back onto the dehazed image via ROI-based alpha blending.

This approach is computationally efficient due to the single-pass generator and ROI-driven blending, making it faster than the Diffusion pipeline.

Diffusion Approach The Diffusion-based pipeline operates as follows:

1. Dehazing is performed using the EDiffSR model with 100 reverse steps.
2. The dehazed output is segmented with U-Net ResNet34 to extract ROI masks.
3. gCNR enhancement is applied to ROI-2 (ventricle), preserving ROI-1 (ventricle septum).
4. ROI-1 is merged back onto the dehazed image via ROI-based alpha blending.
5. Additional blending is applied between the upper and lower parts of the image using a vertical seam and 40-pixel feathering.

While the Diffusion approach provides more gradual refinement and higher-quality restoration, it is computationally more expensive due to the iterative denoising process. Figure 1 visualizes these approaches.

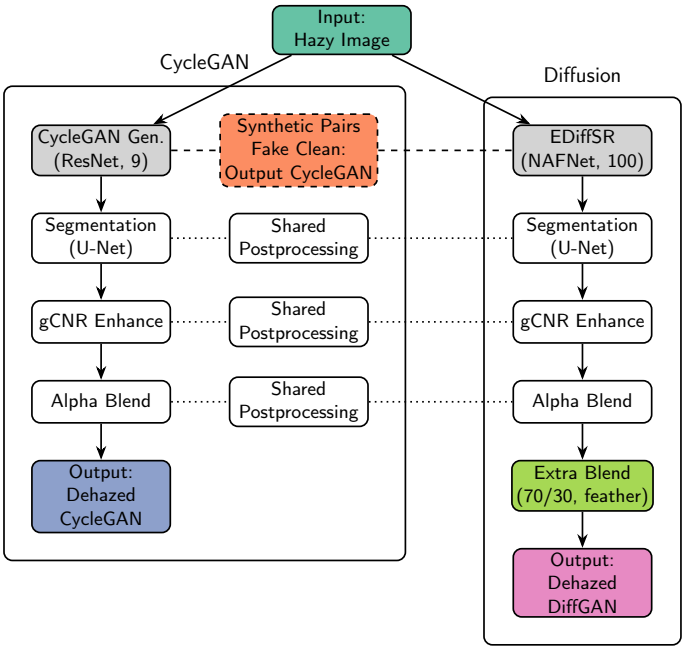


Fig. 1: Overview of CycleGAN and Diffusion dehazing pipelines.

4 Results

The efficacy of the proposed dehazing approaches was evaluated according to the DehazingEcho2025 challenge [1], which employs seven quantitative metrics synthesized into a weighted final score. The evaluation considers three aspects: denoising performance (50%), measured by Fréchet Inception Distance (FID), Contrast-to-Noise Ratio (CNR), and generalized CNR (gCNR) on annotated ROIs (ventricular septum and ventricle); cardiac structure preservation (30%), assessed with the Kolmogorov–Smirnov (KS) test by comparing hazy and dehazed ROIs – low KS values in the septum indicate tissue structure is retained, while high KS values in the ventricle indicate haze removal; and impact on downstream tasks (20%), quantified by Dice coefficient and Average Surface Distance (ASD) of left ventricle segmentation using a pre-trained ultrasound foundation model (USFM) [3]. Figure 2 presents the aggregated scores, demonstrating that all proposed methods substantially improve upon the hazy input, with DiffGAN achieving the highest overall score under the defined weighting scheme. It should be noted that the figures shown below represent only a selection of the seven challenge metrics; others, such as Dice and CNR, also contribute to the final ranking. Figure 3 shows that all methods reduce FID compared to the hazy input, confirming that the dehazed outputs are perceptually closer to clean echocardiograms. The Enhanced CycleGAN achieves the lowest FID, suggesting a particularly realistic global appearance. Figure 4 and Figure 5 further highlight that both

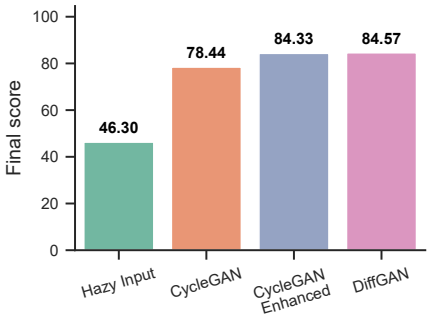


Fig. 2: Final scores comparing all methods according to weighting scheme of DehazingEcho2025 challenge [1].

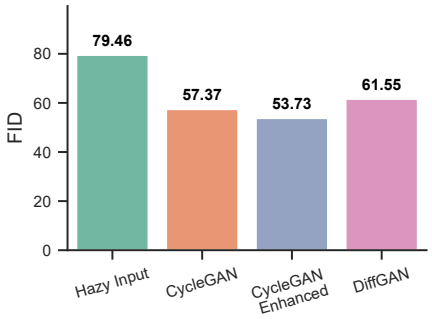


Fig. 3: Fréchet Inception Distance (FID). Lower values indicate closer similarity to clean echocardiograms.

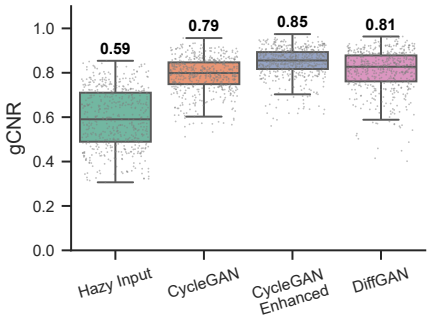


Fig. 4: Generalized Contrast-to-Noise Ratio (gCNR). Higher values indicate improved separability between septum and ventricle.

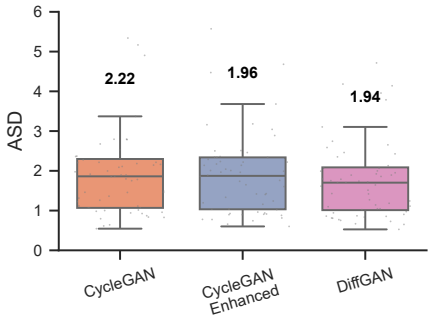


Fig. 5: Average Surface Distance (ASD) of left ventricle segmentation. Lower values indicate more accurate delineation of ventricular boundaries.

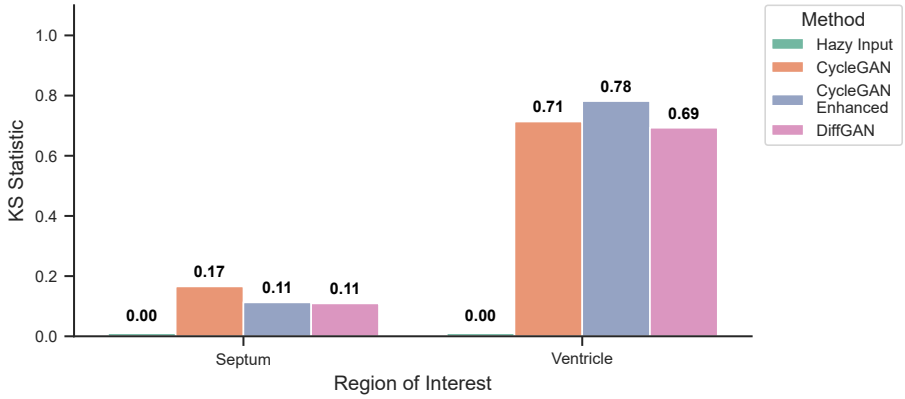
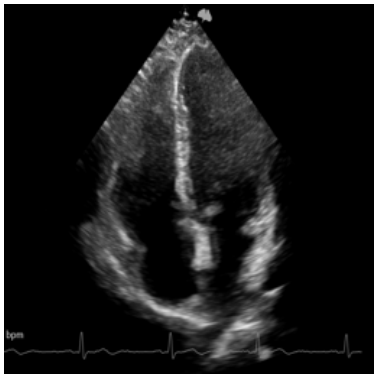


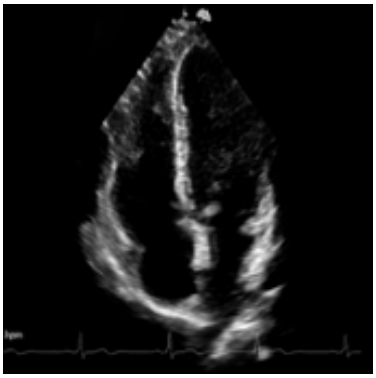
Fig. 6: Kolmogorov–Smirnov (KS) statistic between hazy and dehazed distributions for septum and ventricle. Low KS values in the septum indicate structure preservation; high KS values in the ventricle indicate successful haze removal.

contrast restoration and segmentation consistency benefit from dehazing. The gCNR comparison shows improved separability between septum and ventricle, enhancing diagnostic interpretability. Meanwhile, the ASD plot focuses on the boundary accuracy of downstream left ventricle segmentation. Dice scores were consistently high across all methods (CycleGAN 0.89; CycleGAN Enhanced and DiffGAN 0.90), indicating robust overall segmentation, while ASD provides a more fine-grained view of boundary precision. DiffGAN achieves the lowest ASD, which may contribute to its slightly higher overall score, though the differences between methods remain small. Figure 6 shows the KS statistics. All methods achieve low KS values in the septum, consistent with structural preservation, and high KS values in the ventricle, indicating effective removal of haze. This pattern demonstrates that noise is removed where desired without distorting too much of the key cardiac structures.

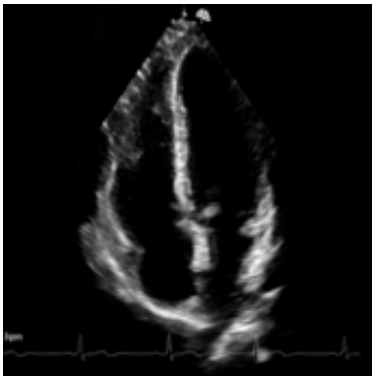
Taken together, the results show complementary strengths of the three approaches: the CycleGAN already achieves strong baseline performance, the Enhanced CycleGAN improves perceptual realism and contrast, and the DiffGAN achieves slightly better overall performance with modest improvement on downstream segmentation. Lastly, Figures 7a to 7c show some qualitative dehazing results.



(a) Hard-to-image patient from DehazingEcho2025 dataset [1].



(b) Dehazed patient with CycleGAN.



(c) Dehazed patient with CycleGAN Enhanced.

Fig. 7: Qualitative comparison of hazy input and dehazed outputs.

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